DATA 430 Technical Report Assignment 1 (a & b): Logistic Regression	Matthew Kolonich
Stroke Prediction Model	
URL to dataset: Stroke Prediction Dataset	

Assignment 1a (due Week 2): you should complete the following sections ONLY:

- Overview (Problem Domain)
- Overview (Objective)
- Analysis (Exploratory Analysis)

Assignment 1b (due Week 3): all sections of this template should be completed. Modifications of the three sections submitted in Assignment 1a should be made based on feedback from the instructor.

This template should be used in conjunction with the assignment instructions. The size of the text area below will expand to the length of your response; the area should not be interpreted as a required or suggested length of response. Responses within the text area should be single spaced with Times New Roman 12pt font. The body of the document will likely be 6-9 pages, not including the Appendix; length may vary depending on specifics of the analysis and the dataset. As needed, APA format in-text citations should be included, along with a full references list at the end of the document.

Overview

Problem Domain: give some background and context about the problem domain (application area). For instance, if you are doing the analysis for predicting heart disease, provide some context about the disease and include some interesting statistics about it. Also, discuss how the method is relevant for the chosen problem.

Stroke prediction is an important area of study within the healthcare industry. Strokes remain one of the leading causes of death and long term disability worldwide. According to the Centers for Disease Control and Prevention (CDC, 2024), a stroke occurs when blood flow to the brain is blocked or when a blood vessel bursts. This prevents brain tissue from receiving oxygen. Early identification of individuals who may be at higher risk for stroke can improve preventative care, reduce medical costs, and potentially save lives. Healthcare organizations increasingly rely on data analytics and machine learning techniques to help identify high risk patients. This help to enable preventive care before severe medical events occur. As healthcare datasets continue to grow in size and complexity, predictive modeling has become an important tool for improving clinical decision making and patient outcomes.

Machine learning is especially valuable in healthcare. It can help identify patterns and relationships within large datasets. These datasets may not be immediately obvious to healthcare professionals. Kubat (2017) explained that machine learning systems learn from pre classified examples in order to make predictions about future outcomes. In supervised learning, models are trained using labeled data. They can then classify or predict future observations based on patterns identified during training. Stroke prediction is considered a supervised learning classification problem because the dependent variable has two possible outcomes: whether a patient experiences a stroke or does not experience a stroke. Logistic regression is well suited for this type of analysis because it is designed to estimate probabilities for binary outcomes. It is also suited because it helps to determine how different variables contribute to the likelihood of a specific event occurring.

This study uses a healthcare stroke prediction dataset containing demographic, lifestyle, and medical information about patients. Variables in the dataset include age, hypertension, heart disease, average glucose level, body mass index (BMI), smoking status, marital status, and work type. These variables may influence an individual's risk of stroke. These variables can therefore be used as independent variables within a predictive model. The dependent variable in the dataset is the occurrence of stroke. By analyzing the relationships, machine learning models can assist healthcare professionals in identifying high risk patients. They can also improve preventative intervention strategies. In addition, predictive analytics can help healthcare systems allocate resources more effectively and support earlier treatment decisions for vulnerable populations.

Objective: clearly state the objective of the analysis in relation to the kind of algorithm you are employing. Use specific language as to what question(s) you are trying to answer using the specific analysis/modeling type.

The primary objective of this study is to develop and evaluate a logistic regression model capable of predicting whether a patient is likely to experience a stroke. The prediction will be based on demographic, lifestyle, and medical characteristics contained within the dataset. The study focuses on applying supervised machine learning techniques to identify relationships between patient attributes and stroke occurrence. Since stroke prediction involves two possible outcomes logistic regression is an appropriate modeling technique for this classification problem. The model will estimate the probability that a patient belongs to the positive stroke class based on the independent variables provided in the dataset.

Another objective of this study is to explore how different healthcare variables may contribute to stroke risk. Variables such as age, hypertension, heart disease, average glucose level, body mass index (BMI), and smoking status are expected to influence the likelihood of stroke occurrence. Through exploratory analysis and preprocessing, the study will identify important patterns, distributions, and relationships within the data before fitting the logistic regression model. Understanding these relationships is important. Healthcare datasets often contain missing values, categorical variables, and imbalanced class distributions that can affect model performance and prediction accuracy.

This study also aims to demonstrate the practical use of machine learning within the healthcare industry. According to Kubat (2017), classification models are designed to learn patterns from labeled training examples and apply those learned relationships to future observations. Logistic regression provides a useful foundation for predictive healthcare analytics. It offers interpretable results while still producing effective classification performance. Unlike some more complex machine learning algorithms that operate as black box systems, logistic regression allows researchers and healthcare professionals to better understand how specific variables contribute to predicted outcomes.

Also, this study seeks to evaluate how preprocessing techniques can improve model performance. These techniques include handling missing values, encoding categorical variables, and preparing independent variables. The project will also provide practical experience with exploratory data analysis, data visualization, and classification modeling using Python libraries. These libraries include Pandas, Matplotlib, Seaborn, and Scikit learn. By completing this analysis, the study aims to demonstrate how machine learning can support healthcare decision making and assist in the early identification of patients who may be at increased risk for stroke.

Analysis

Exploratory Analysis: describe the data including the source, the collection method, and variables. Perform exploratory analysis. Also, select few key variables (including the target variable for supervised learning) and study their distributions using plots such as histograms, box plot, bar chart, etc.

The dataset used for this study is the Stroke Prediction Dataset from Kaggle. The dataset contains 5,110 patient records and 12 original columns. The columns include demographic, medical, and lifestyle variables. The variables include id, gender, age, hypertension, heart_disease, ever_married, work_type, Residence_type, avg_glucose_level, bmi, smoking_status, and stroke. The dependent variable for this supervised learning classification problem is stroke. Stoke is the variable which identifies whether the patient had a stroke. The independent variables include age, hypertension, heart disease, average glucose level, BMI, smoking status, work type, residence type, gender, and marital status. The id field was excluded from the model because it is only an identifier and does not provide meaningful predictive value.

The exploratory analysis began by loading the required Python libraries. These included NumPy, Pandas, Matplotlib, Seaborn, Scikit learn's train test split function, LogisticRegression, StandardScaler, and model evaluation tools. After loading the dataset, I reviewed the structure of the data using info(). I found that the dataset contained both numeric and categorical variables. Numeric variables included age, hypertension, heart disease, average glucose level, BMI, and stroke. Categorical variables included gender, ever married, work type, residence type, and smoking status. Since logistic regression requires numeric input, the categorical variables were converted into dummy variables using one hot encoding.

The missing value check showed that BMI was the only variable with missing values. There were 201 BMI records that were missing out of 5,110 total observations. To avoid dropping a large number of records, the missing BMI values were replaced with the mean BMI value. This allows the dataset to keep all patient records while still preparing the variable for analysis. I also reviewed the descriptive statistics to better understand the numeric variables. The mean age was approximately 43.23 years, while the average BMI was approximately 28.89. The average glucose level was approximately 106.15, but the maximum value was 271.74. This suggests some patients had very high glucose levels.

One of the most important findings from the descriptive statistics was the imbalance in the target variable. The mean value of the stroke variable was approximately 0.049. This means only about 4.9% of the dataset represented stroke cases. This is important because a model could appear accurate by mostly predicting the majority class, which is no stroke. From a healthcare perspective this really matters because the main goal is not simply to predict the most common outcome. The more meaningful goal is to identify patients who may actually be at risk of stroke.

The age histogram showed that the dataset includes patients across a wide age range. Patients range from very young individuals to patients in their eighties. However, the distribution showed a noticeable number of observations among middle aged and older patients. This is relevant because age is commonly associated with stroke risk. This may become one of the stronger predictors in the logistic regression model. The BMI histogram showed that most BMI values were clustered near the overweight range with some high BMI outliers. The Stroke vs. BMI bar plot suggested that BMI may have some relationship with stroke occurrence. Though, the graph should be interpreted carefully because BMI contains many individual numeric values.

After the exploratory analysis, the dataset was split into training and testing sets using a 75% training and 25% testing split. The split was stratified by the target variable to preserve the proportion of stroke and non-stroke cases in both sets. This was important because the dataset is highly imbalanced. The independent variables were then standardized using StandardScaler. This helped place numeric variables on a comparable scale before fitting the logistic regression model.

The updated logistic regression model used balanced class weighting in order to address the highly imbalanced nature of the dataset. Initially, the model achieved a very high accuracy score of approximately 95.2%, but it failed to correctly identify most stroke cases. After applying class balancing techniques, the updated model achieved an accuracy score of approximately 72.8%. Although the overall accuracy decreased, the recall score for stroke cases improved significantly from 0.02 to 0.79. This means the updated model successfully identified most of the actual stroke cases within the testing data.

The confusion matrix showed that the model correctly identified 49 out of 62 stroke cases, while also generating a larger number of false positives. Specifically, the model correctly predicted 882 non-stroke cases and 49 stroke cases. It incorrectly classified 334 non-stroke cases as strokes and missing 13 actual stroke cases. This reflects an important tradeoff in healthcare machine learning applications. In medical prediction tasks, false negatives can be especially dangerous because failing to identify a high risk patient may delay treatment or preventative intervention. From a healthcare perspective, improving recall is often more valuable than simply maximizing overall accuracy.

The classification report further demonstrated the strengths and limitations of the updated model. The recall score for stroke cases improved dramatically to 0.79, indicating that the model successfully identified most stroke patients in the testing set. However, the precision score for stroke prediction was only 0.13, showing that many predicted stroke cases were false positives. The F1-score for stroke prediction was 0.22, reflecting the balance between recall and precision. While the model generated more false alarms, it became substantially more effective at identifying patients who may actually be at risk of stroke.

Personally, this portion of the project helped me better understand how misleading accuracy alone can be in classification problems involving imbalanced healthcare data. A model can appear highly accurate while still failing to identify the minority class that matters most. This reinforced the importance of evaluating machine learning models using additional metrics such as recall, precision, F1-score, sensitivity, specificity, and confusion matrices. It is better than relying only on overall accuracy. From a healthcare perspective, identifying high risk patients early may be more valuable than simply maximizing prediction accuracy across the majority class.

The age distribution histogram shows that the dataset includes patients across a wide age range. There is a larger concentration among middle aged and older adults. This is important because age is commonly associated with increased stroke risk. The graph suggests that age may become one of the stronger predictive variables within the logistic regression model.

The Stroke vs. BMI graph suggests there may be a relationship between BMI and stroke occurrence. Higher BMI values appear across several stroke observations. This may indicate that body weight and overall health conditions could contribute to stroke risk. However, because BMI contains many unique numeric values, the graph should be interpreted cautiously. It should be viewed as an exploratory visualization rather than definitive proof of correlation.

The BMI distribution histogram shows that most patients fall within the overweight BMI range. There are several higher value outliers present in the dataset. This indicates variation in patient health profiles. It also suggests that BMI may provide useful information during model training. The presence of outliers also highlights the importance of preprocessing and scaling before fitting the logistic regression model.

Preprocessing: armed with the exploratory analysis, perform the necessary preprocessing, both general and specific types appropriate for the modeling type being employed.

Prior to fitting the logistic regression model, several preprocessing steps were completed. This was to improve data quality and ensure that the dataset was suitable for machine learning analysis. Data preprocessing is an important stage in the machine learning lifecycle because model performance is heavily influenced by the quality and preparation of the underlying data. As noted by GeeksforGeeks (2024), data preparation and preprocessing are critical components of the machine learning process because they help ensure that models are trained using clean, consistent, and meaningful data.

The first preprocessing step involved examining the dataset for missing values. A review of the dataset identified 201 missing values within the Body Mass Index (BMI) variable. Since BMI is a potentially important predictor of stroke risk, removing all records with missing BMI values would have unnecessarily reduced the size of the dataset. Instead, the missing BMI values were replaced using the mean BMI value calculated from the available observations. This approach preserved the sample size while maintaining consistency within the dataset.

The next preprocessing step involved converting categorical variables. They needed to be converted into a format suitable for logistic regression analysis. Several variables in the dataset, including gender, marital status, work type, residence type, and smoking status, were stored as text values. Because logistic regression requires numerical inputs, one-hot encoding was applied using the pandas get_dummies() function. This process transformed categorical values into binary indicator variables while using the drop_first=True parameter to reduce redundancy and minimize multicollinearity.

After the categorical variables were encoded, the independent and dependent variables were defined. The dependent variable was stroke. This represents whether a patient experienced a stroke. The independent variables included age, hypertension, heart disease, average glucose level, BMI, and all encoded categorical variables. The unique patient identification number was removed because it does not contribute meaningful predictive information and could introduce noise into the model.

The dataset was then divided into training and testing datasets using a 75% training and 25% testing split. Stratified sampling was applied to preserve the proportion of stroke and non-stroke cases in both datasets. This step was particularly important because the dataset is highly imbalanced. The stroke cases represented only a small percentage of the overall observations. Maintaining class proportions helps ensure that the evaluation results accurately reflect real-world performance.

Feature scaling was also performed using StandardScaler from scikit-learn. Logistic regression models often perform better when numerical variables are standardized. This is because variables measured on different scales can disproportionately influence the model coefficients. Standardization transformed the numerical predictors to have a mean of zero and a standard deviation of one. They also preserved the underlying relationships among variables.

As I worked through the preprocessing stage, I gained a greater appreciation for how much effort occurs before a model is ever trained. Prior to this project, I assumed that building a machine learning model primarily involved selecting an algorithm and generating predictions. However, this analysis demonstrated that preparing the data is one of the most important aspects of the machine learning process. Decisions regarding missing values, categorical encoding, data partitioning, and feature scaling all have a direct impact on the quality and reliability of the final model. These preprocessing steps established a solid foundation for the logistic regression model and improved confidence in the results produced during the subsequent model fitting and evaluation phases.

Model Fitting: explain the key steps and activities you perform to fit the model. Experiment (as appropriate) with parameters tuning. This is key, what separates highly accurate model from a less accurate ones is the amount of performance tuning performed.

After I completed the preprocessing stage, a logistic regression model was developed to predict whether a patient was likely to experience a stroke. Logistic regression was selected because the dependent variable, stroke, is binary and can only take one of two values: stroke or no stroke. According to Kubat (2017), classification algorithms are designed to learn relationships from labeled examples and apply those learned patterns to future observations. Logistic regression is one of the most widely used classification techniques. This is because it provides a relatively simple and interpretable approach to predicting categorical outcomes.

The dataset was divided into training and testing datasets using a 75% training and 25% testing split. The training dataset was used to fit the model. The testing dataset was reserved for evaluating performance on unseen data. Stratified sampling was applied during the split to ensure that the proportion of stroke and non-stroke cases remained consistent across both datasets. This was especially important because stroke cases represented less than 5% of all observations.

Before I fitted the model, the independent variables were standardized using the StandardScaler function from scikit-learn. Standardization transformed the numerical variables to a common scale with a mean of zero and a standard deviation of one. This preprocessing step helps prevent variables with larger numerical ranges from exerting excessive influence on the model coefficients. It can improve optimization during model training.

The initial logistic regression model was trained using the default parameter settings and a maximum iteration value of 1,000. However, the results revealed a common problem associated with highly imbalanced datasets. Although the model achieved a high overall accuracy score, it failed to correctly identify most stroke cases. The model primarily predicted the majority class. This resulted in a very low recall for the minority stroke class. This demonstrated that accuracy alone was not an adequate measure of model effectiveness for this healthcare application.

To address this limitation, the logistic regression model was modified using the class_weight='balanced' parameter. This parameter automatically adjusts the importance assigned to each class during model training. Because stroke cases are relatively rare within the dataset, the balanced weighting approach places greater emphasis on correctly identifying stroke patients. This adjustment helps reduce bias toward the majority class. It also improves the model's ability to detect high-risk individuals.

The final logistic regression model therefore included two key parameters: max_iter=1000 and class_weight='balanced'. The increased iteration limit ensured that the optimization process had sufficient opportunities to converge. Balanced class weighting also improved the model's sensitivity to stroke cases. The model was then trained using the scaled training data and evaluated using the testing dataset.

One aspect of this project that I found particularly interesting was observing how a small parameter adjustment could significantly change model behavior. Initially, the model appeared highly accurate but performed poorly on the outcome that mattered most. After introducing class balancing, overall accuracy decreased, but the model became substantially better at identifying actual stroke cases. This experience reinforced the importance of understanding both the dataset and the evaluation metrics when developing machine learning models. It also demonstrated that selecting an appropriate model configuration is often just as important as selecting the algorithm itself.

As Rajkomar et al. (2019) explain, machine learning models have the potential to transform healthcare data into actionable clinical insights. The goal of this project was not simply to maximize accuracy but to develop a model capable of identifying patients who may be at elevated risk of stroke. The model fitting process therefore focused on balancing predictive performance with practical healthcare considerations.

Results

Model Properties: explain the components of the fitted model and their characteristics. Leverage functions to summarize the model properties. Also, leverage visualization as required.

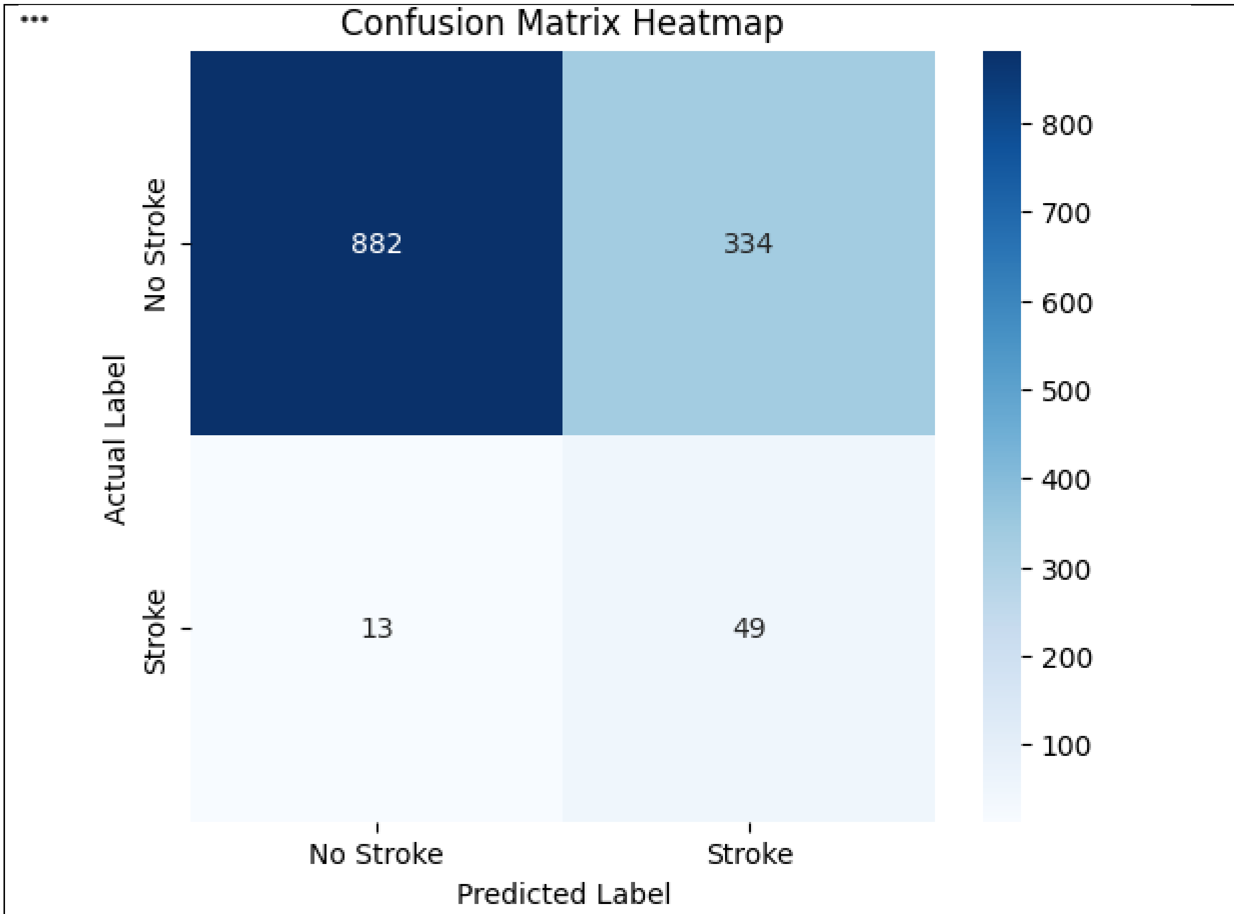
A logistic regression model was developed to predict whether a patient experienced a stroke using demographic, lifestyle, and health-related variables. The model was trained using a dataset containing 5,110 patient records and multiple predictor variables. These included age, hypertension, heart disease, average glucose level, body mass index (BMI), smoking status, work type, marital status, residence type, and gender. Prior to model fitting, categorical variables were transformed into numerical dummy variables and continuous variables were standardized using StandardScaler.

To address the class imbalance present within the dataset, the Logistic Regression algorithm was configured with the parameter `class_weight='balanced'`. This adjustment was necessary because only approximately 4.9% of observations represented stroke cases, while the remaining observations represented non-stroke cases. Without this adjustment, the model would tend to predict the majority class and fail to identify patients at risk for stroke.

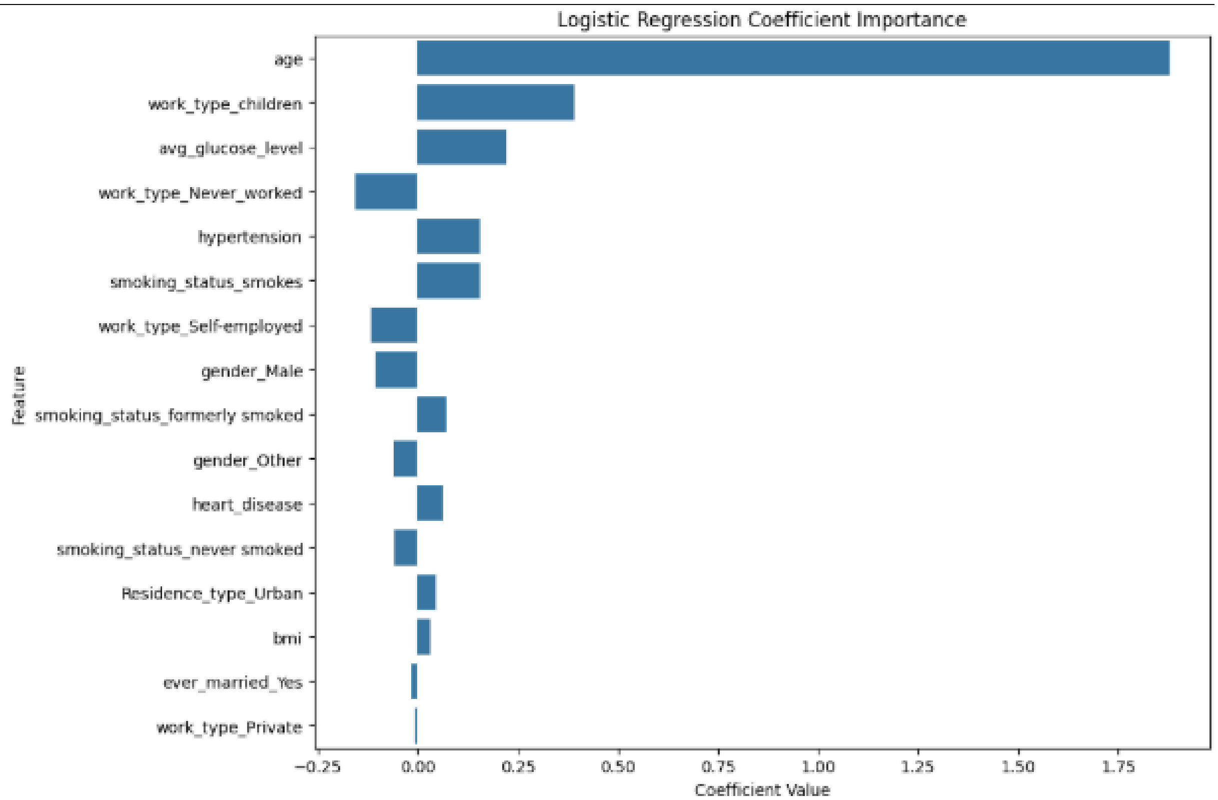
The model was trained using a 75% training sample and evaluated using the remaining 25% testing sample. The Logistic Regression algorithm converged successfully after setting the maximum iteration parameter to 1,000. Model evaluation included analysis of the confusion matrix, classification report, accuracy score, and coefficient importance values.

The resulting model achieved an overall accuracy of 72.85%. This accuracy is lower than the initial model developed without class balancing. It provides a more realistic representation of predictive performance because the model is capable of identifying a substantially larger number of stroke cases. The model achieved a recall value of 0.79 for stroke cases. This indicates that approximately 79% of actual stroke events were correctly identified. This metric is particularly important in healthcare applications where missing a potential stroke patient may have serious consequences.

The coefficient analysis revealed that age was the strongest predictor of stroke occurrence. This was followed by work type, average glucose level, hypertension, and smoking status. These findings are generally consistent with established medical research regarding stroke risk factors. This provides evidence that the model is identifying meaningful relationships within the dataset.



The confusion matrix demonstrates that the model correctly classified 882 non-stroke cases and 49 stroke cases. The model also incorrectly classified 334 non-stroke cases as strokes and failed to identify 13 stroke cases. Although the number of false positives is relatively high, the model successfully identifies the majority of stroke cases. This is preferred in healthcare screening applications where patient safety is a primary concern.



The coefficient chart indicates that age is the most influential predictor of stroke risk within the model. Additional influential variables include work type, average glucose level, hypertension, and smoking status. Positive coefficient values indicate increased stroke risk. The negative values indicate variables associated with reduced risk within the model.

Output Interpretation: explain the result and interpret the final model output using terms that reflect the application area and in relation to the stated objective. This is where you check whether or not the stated objective is met.

The logistic regression model achieved an overall accuracy of 72.85%. This means that approximately 73% of observations in the testing dataset were classified correctly. While accuracy is commonly used as a performance metric, it should not be viewed in isolation. This is because the stroke dataset is highly imbalanced, with only a small percentage of observations representing actual stroke cases. In healthcare applications, relying solely on accuracy can be misleading. This is because a model may achieve high accuracy while failing to identify patients who are truly at risk.

The confusion matrix showed that the model correctly identified 882 non-stroke cases and 49 stroke cases. In addition, the model incorrectly classified 334 non-stroke cases as strokes and failed to identify 13 stroke cases. These results indicate that the model favors identifying potential stroke patients even at the expense of producing more false positive predictions.

The classification report provides additional insight into model performance. For the stroke class, the model achieved a precision score of 0.13, a recall score of 0.79, and an F1-score of 0.22. The relatively low precision indicates that many patients identified as high risk did not actually experience a stroke. However, the recall value of 0.79 is particularly important because it demonstrates that the model successfully detected approximately 79% of actual stroke cases.

From a healthcare perspective the recall is often more important than precision. This is because the consequences of failing to identify a patient at risk for stroke can be severe. A false positive prediction may result in additional testing or monitoring, while a false negative prediction could prevent a patient from receiving early intervention. For this reason, the balanced logistic regression model provides a more practical solution than the original model, which achieved higher accuracy but failed to identify most stroke cases.

The coefficient importance analysis further supports the model's validity. Age emerged as the strongest predictor of stroke risk. This was followed by variables such as average glucose level, hypertension, smoking status, and work type. These findings align with established medical research. They suggest that the model is capturing meaningful relationships within the dataset rather than producing random predictions.

Personally, I found this portion of the project particularly interesting. It demonstrated how evaluation metrics can tell very different stories about model performance. Before completing this analysis, I would have assumed that a model with higher accuracy was automatically better. However, comparing the original model to the balanced model showed that accuracy alone does not determine usefulness. In a healthcare setting, identifying patients who may be at risk is often more valuable than achieving the highest possible accuracy score. This reinforced the importance of selecting evaluation metrics that align with the real-world objectives of the problem being studied.

Evaluation: employ appropriate metrics to quantitatively evaluate the performance of the fitted model. For supervised classification, this includes simple accuracy, precision & recall (or sensitivity & specificity), all of which can be generated from a confusion matrix, or ROC.

The performance of the logistic regression model was evaluated using several commonly accepted machine learning metrics. They included accuracy, precision, recall, F1-score, and the confusion matrix. Evaluating multiple metrics is particularly important when working with imbalanced datasets. This is because a single metric may not provide a complete picture of model performance. Kubat (2017) emphasizes that classification accuracy alone can be misleading in domains where one class is significantly underrepresented, as a model may achieve high accuracy while failing to correctly identify important minority cases.

The logistic regression model achieved an overall classification accuracy of 72.85%. While this accuracy is lower than the original model developed without class balancing, it represents a more realistic assessment of predictive performance. This is because the model successfully identified a substantially greater number of stroke cases. Given that stroke occurrences represent less than 5% of the observations in the dataset, overall accuracy is less informative than metrics that evaluate performance on the minority class.

The confusion matrix showed that the model correctly classified 882 non-stroke cases and 49 stroke cases. The model also incorrectly classified 334 non-stroke cases as strokes and failed to identify 13 stroke cases. These results demonstrate that the model prioritizes identifying potential stroke patients, even if doing so increases the number of false positive predictions. In healthcare settings, this tradeoff is often acceptable. This is because the consequences of missing a patient who may experience a stroke are generally more severe than performing additional follow-up evaluations on patients who ultimately do not experience one.

The classification report revealed a precision score of 0.13, a recall score of 0.79, and an F1-score of 0.22 for the stroke class. Although the precision score is relatively low, the recall score indicates that the model successfully identified approximately 79% of actual stroke cases. This result is particularly significant because recall measures the model's ability to detect positive cases. This is often the primary objective in medical screening applications. According to Kubat (2017), recall becomes especially important when the cost of failing to identify positive examples is high.

The coefficient importance analysis provided additional insight into the factors influencing model predictions. Age emerged as the strongest predictor of stroke risk. Age was followed by work type, average glucose level, hypertension, and smoking status. These findings align with established medical research identifying age, cardiovascular conditions, elevated glucose levels, and smoking behaviors as significant stroke risk factors. The consistency between the model's findings and known clinical risk factors increases confidence that the model is identifying meaningful patterns within the data.

Overall, the logistic regression model demonstrated moderate predictive performance. It also successfully addressed the class imbalance problem through the use of balanced class weights. Although more advanced machine learning techniques such as Random Forest, Gradient Boosting, or XGBoost may improve predictive performance, the logistic regression model provides an interpretable and effective baseline solution. From my perspective, one of the most valuable lessons from this evaluation was learning that a model with lower accuracy can actually be more useful when it better identifies the cases that matter most. This project reinforced the importance of selecting evaluation metrics that align with the real-world goals of the problem rather than relying exclusively on overall accuracy.

Conclusion

Summary: highlight the main findings in relation to the stated objective. You don't need to discuss the details of the analysis and the model such as accuracy here, just focus on the key findings.

The purpose of this study was to develop and evaluate a logistic regression model capable of predicting stroke occurrence using demographic, lifestyle, and health-related variables from a healthcare dataset. The project followed a structured machine learning process. This included exploratory data analysis, data preprocessing, model fitting, and performance evaluation. Several potential risk factors were examined. These included age, hypertension, heart disease, average glucose level, body mass index (BMI), smoking status, work type, and marital status. Through this process, the study demonstrated how machine learning techniques can be applied to real-world healthcare problems. It provided meaningful insights into the factors associated with stroke risk.

The exploratory analysis revealed several important characteristics of the dataset. It also highlighted the challenges associated with healthcare data. Missing BMI values required preprocessing before model development. Categorical variables had to be transformed into numerical representations through one-hot encoding. Feature scaling was also necessary to ensure that variables measured on different scales could be appropriately evaluated by the logistic regression algorithm. These preprocessing steps improved the quality of the dataset and prepared it for effective model training.

The final logistic regression model achieved an overall accuracy of 72.85% and successfully identified approximately 79% of actual stroke cases. Although the model generated a relatively large number of false positive predictions, it demonstrated strong recall performance. This is particularly valuable in healthcare applications where failing to identify a patient at risk can have serious consequences. The coefficient analysis identified age as the most influential predictor of stroke occurrence, followed by factors such as average glucose level, hypertension, smoking status, and work type. These findings are consistent with established medical knowledge regarding stroke risk factors.

One of the most important lessons I learned during this project was that accuracy alone does not necessarily indicate a model's usefulness. At the beginning of the project, I assumed that the model with the highest accuracy would automatically be the best model. However, after evaluating the balanced logistic regression model, I learned that other metrics are factors. These metrics include recall, precision, and F1-score. They can provide a more complete understanding of model performance. In this case, a model with lower overall accuracy proved to be more effective at identifying stroke cases. This is ultimately the primary objective of the study.

Another valuable lesson was the importance of data preparation and preprocessing. The quality of the final model depended heavily on properly handling missing values, transforming categorical variables, and scaling numerical features. This project reinforced the idea that successful machine learning involves much more than simply applying an algorithm to a dataset. Data preparation often plays a critical role in determining the quality of the final results.

According to Hosmer, Lemeshow, and Sturdivant (2013), logistic regression is particularly valuable because it provides both predictive capability and interpretability, allowing researchers to better understand how individual variables influence outcomes. This project demonstrated that advantage firsthand by identifying meaningful predictors of stroke risk while maintaining a model that is relatively easy to interpret and explain. Overall, this study provided practical experience with the complete machine learning process. It strengthened my understanding of how data preparation, model selection, and performance evaluation work together to solve real-world problems.

Limitations & Improvement areas: discuss the limitations of the analysis and identify potential improvement areas for future work. This could be related to the data, algorithm, or a combination of the two.

While the logistic regression model produced useful results, several limitations should be acknowledged. The most significant limitation is the highly imbalanced nature of the dataset. Only a small percentage of observations represented actual stroke cases. This makes it difficult for the model to learn patterns associated with the minority class. Although the use of balanced class weights improved the model's ability to identify stroke cases, the model still produced a relatively high number of false positive predictions. This tradeoff illustrates the challenges associated with predicting rare medical events.

Another limitation is the scope of the available data. The dataset included several important health and demographic variables. Though, it did not contain potentially influential factors such as family medical history, physical activity levels, dietary habits, medication usage, cholesterol measurements, or genetic information. The absence of these variables may limit the model's ability to fully capture the complex factors that contribute to stroke risk. As discussed throughout this course, machine learning models can only learn from the information that is available within the dataset.

A further limitation involves the use of mean imputation to replace missing BMI values. While this approach allowed all observations to remain in the analysis, it may have reduced some of the natural variation within the dataset. Alternative techniques such as median imputation, K-nearest neighbors imputation, or multiple imputation methods may preserve additional information and potentially improve model performance.

The logistic regression algorithm itself also introduced limitations. Although logistic regression provides strong interpretability and is relatively easy to explain, it assumes a linear relationship between predictor variables and the log-odds of the outcome. Real-world healthcare relationships are often more complex. They may involve nonlinear interactions that logistic regression cannot fully capture. More advanced machine learning algorithms may be better suited for modeling these complex relationships.

Several improvement opportunities exist for future research. Additional feature engineering could be performed to create new variables that better capture patient risk factors. Future studies could also explore alternative machine learning techniques such as Random Forest, Gradient Boosting, XGBoost, or Neural Networks. These could help to determine whether predictive performance can be improved. Comparing multiple algorithms on the same dataset would provide a better understanding of which approach is most appropriate for stroke prediction.

One improvement I would make if continuing this project would be to experiment with additional methods for handling class imbalance, such as SMOTE (Synthetic Minority Oversampling Technique) or under sampling techniques. I would also investigate whether additional healthcare variables could be incorporated into the dataset. Through this project, I learned that model performance is often influenced as much by data quality and preparation as by the machine learning algorithm itself. Future work focused on improving both the dataset and the modeling approach could lead to more accurate and reliable stroke prediction systems.

Overall, while the logistic regression model demonstrated practical value as an interpretable baseline solution, there remains considerable opportunity for improvement. This can be achieved through enhanced data collection, feature engineering, and the exploration of more sophisticated machine learning techniques.

Appendix

```
      id  gender   age  hypertension  heart_disease ever_married  \
0   9046    Male  67.0             0              1          Yes   
1  51676  Female  61.0             0              0          Yes   
2  31112    Male  80.0             0              1          Yes   
3  60182  Female  49.0             0              0          Yes   
4   1665  Female  79.0             1              0          Yes   

       work_type Residence_type  avg_glucose_level   bmi   smoking_status  \
0        Private          Urban             228.69  36.6  formerly smoked   
1  Self-employed          Rural             202.21   NaN     never smoked   
2        Private          Rural             105.92  32.5     never smoked   
3        Private          Urban             171.23  34.4           smokes   
4  Self-employed          Rural             174.12  24.0     never smoked   

   stroke  
0       1  
1       1  
2       1  
3       1  
4       1  
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5110 entries, 0 to 5109
Data columns (total 12 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   id                 5110 non-null   int64  
 1   gender             5110 non-null   object 
 2   age                5110 non-null   float64
 3   hypertension       5110 non-null   int64  
 4   heart_disease      5110 non-null   int64  
 5   ever_married       5110 non-null   object 
 6   work_type          5110 non-null   object 
 7   Residence_type     5110 non-null   object 
 8   avg_glucose_level  5110 non-null   float64
 9   bmi                4909 non-null   float64
 10  smoking_status     5110 non-null   object 
 11  stroke             5110 non-null   int64  
dtypes: float64(3), int64(4), object(5)
memory usage: 479.2+ KB
None
id                     0
gender                 0
age                    0
```

```
hypertension           0
heart_disease          0
ever_married           0
work_type              0
Residence_type         0
avg_glucose_level      0
bmi                  201
smoking_status         0
stroke                 0
dtype: int64
id                   0
gender               0
age                  0
hypertension         0
heart_disease        0
ever_married         0
work_type            0
Residence_type       0
avg_glucose_level    0
bmi                  0
smoking_status       0
stroke               0
dtype: int64
                 id          age  hypertension  heart_disease  \
count   5110.000000  5110.000000   5110.000000    5110.000000
mean   36517.829354    43.226614      0.097456       0.054012
std    21161.721625    22.612647      0.296607       0.226063
min       67.000000     0.080000      0.000000       0.000000
25%    17741.250000    25.000000      0.000000       0.000000
50%    36932.000000    45.000000      0.000000       0.000000
75%    54682.000000    61.000000      0.000000       0.000000
max    72940.000000    82.000000      1.000000       1.000000

       avg_glucose_level          bmi       stroke
count        5110.000000  5110.000000  5110.000000
mean          106.147677    28.893237     0.048728
std            45.283560     7.698018     0.215320
min            55.120000    10.300000     0.000000
25%            77.245000    23.800000     0.000000
50%            91.885000    28.400000     0.000000
75%           114.090000    32.800000     0.000000
max           271.740000    97.600000     1.000000
      id   age  hypertension  heart_disease  avg_glucose_level        bmi  \
0   9046  67.0             0              1             228.69  36.600000
1  51676  61.0             0              0             202.21  28.893237
2  31112  80.0             0              1             105.92  32.500000
3  60182  49.0             0              0             171.23  34.400000
4   1665  79.0             1              0             174.12  24.000000

   stroke  gender_Male  gender_Other  ever_married_Yes  \
0       1         True         False              True
1       1        False         False              True
2       1         True         False              True
3       1        False         False              True
4       1        False         False              True

   work_type_Never_worked  work_type_Private  work_type_Self-employed  \
```

```
0                   False               True                       False
1                   False              False                        True
2                   False               True                       False
3                   False               True                       False
4                   False              False                        True

   work_type_children  Residence_type_Urban  smoking_status_formerly smoked
\
0               False                  True                            True
1               False                 False                           False
2               False                 False                           False
3               False                  True                           False
4               False                 False                           False

   smoking_status_never smoked  smoking_status_smokes
0                        False                  False
1                         True                  False
2                         True                  False
3                        False                   True
4                         True                  False




Confusion Matrix:
[[882 334]
 [ 13  49]]

Accuracy Score:
0.7284820031298904
Classification Report:
              precision    recall  f1-score   support

           0       0.99      0.73      0.84      1216
           1       0.13      0.79      0.22        62

    accuracy                           0.73      1278
   macro avg       0.56      0.76      0.53      1278
weighted avg       0.94      0.73      0.81      1278

Logistic Regression Coefficients:
                           Feature  Coefficient  Absolute_Coefficient
0                              age     1.878491              1.878491
11              work_type_children     0.392615              0.392615
3                avg_glucose_level     0.219692              0.219692
8           work_type_Never_worked    -0.156511              0.156511
1                     hypertension     0.155445              0.155445
15           smoking_status_smokes     0.154411              0.154411
10         work_type_Self-employed    -0.116556              0.116556
5                      gender_Male    -0.103964              0.103964
13  smoking_status_formerly smoked     0.070178              0.070178
6                     gender_Other    -0.061628              0.061628
2                    heart_disease     0.061462              0.061462
14     smoking_status_never smoked    -0.058055              0.058055
12            Residence_type_Urban     0.043505              0.043505
4                              bmi     0.031312              0.031312
7                 ever_married_Yes    -0.016852              0.016852
```

9	work_type_Private	-0.006361	0.006361

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